POLITECNICO di MILANO 1863

MRN Project

Planning Optimization for Wireless Network Deployment

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Introduction

In today's competitive telecommunications landscape, network operators must balance coverage quality, cost efficiency, and performance guarantees when planning new infrastructure. As demand for wireless services continues to grow, careful network planning and optimization becomes crucial—especially in public or urban environments where both user density and deployment costs are high.

This project addresses a coverage planning problem for a large telecom company seeking to deploy wireless connectivity in a defined geographic area. The company has the option to install macro base stations (high-power, high-cost) or micro base stations (low-power, low-cost) at several candidate sites. Each installation must guarantee a minimum peak throughput of 7 Mbps for all designated test points (TPs) within the area.

The telecom company wants to answer three key questions:

- Is it more cost-effective to use only macro or only micro base stations?
- What is the minimum total cost of a feasible deployment using either option?
- Can the cost be further reduced by allowing a mixed deployment of both base station types?

To answer these questions, we model the deployment as a Mixed-Integer Linear Programming (MILP) problem, incorporating radio propagation, coverage constraints, and throughput requirements. We then evaluate the deployment costs under different scenarios and visualize the resulting network layout.

Sets:

Symbol	Description
C	Set of candidate sites where a base station (BS) can be installed, indexed by c
T	Set of test points that must be covered, indexed by t
B	Set of base station types, B = {macro, micro} indexed by b

Table 1

Parameters:

Symbol	Description
$\cos t_{cb}$	Installation cost of a BS of type b at candidate site c
p_{cbt}^{rx}	Received power (in dBm) at test point t from a BS of type b installed at site c
γ	Receiver sensitivity threshold, the minimum received power for coverage (−75 dBm)
N	Noise power (−95 dBm)
f_b	Bandwidth available to BS type b (20 MHz for macro, 10 MHz for micro)
p_b^{tx}	Transmit power of a BS of type b (40 dBm for macro, 30 dBm for micro)
g_t, g_r	Antenna gains for transmitter and receiver (unitless, typically 1 dB)
Thr_{cbt}	Peak throughput achievable from BS (c, b) to test point t, computed using Shannon's formula
SNR_{cbt}	Signal-to-noise ratio of the link from BS (c, b) to test point t, in linear scale

Table 2

Decision Variables:

Variable	Type	Description
$\boldsymbol{x_{cb}} \in \{\mathbf{0}, \mathbf{1}\}$	Binary	1 if a BS of type b is installed at site c; 0 otherwise
$\boldsymbol{y_{cbt}} \in \{\mathbf{0}, \mathbf{1}\}$	Binary	1 if test point t is assigned to BS (c, b); 0 otherwise

Table 3

These sets, parameters, and variables form the foundation of the MILP model used to optimize the wireless network deployment.

Objective Function:

Minimize the total installation cost of all deployed base stations:

$$\min \sum_{c \in C} \sum_{b \in B} cost_{cb} . x_{cb}$$

Constraints:

1. Assignment constraint

   Each test point must be assigned to exactly one base station:

$$\sum_{c \in C} \sum_{b \in B} y_{cbt} = 1 \qquad \forall t \in T$$

2. Activation Constraint

   A test point can only be assigned to a base station if that base station is installed:

$$y_{cbt} \leq x_{cb} \qquad \forall c \in C, b \in B, t \in T$$

3. Coverage Constraint
   Test points can only be served by BSs where received power is above the receiver sensitivity γ :

$$y_{cbt}.p_{cbt}^{rx} \geq y_{cbt}.\gamma \qquad \forall c, b, t$$

(equivalent to: if $y_{cbt} = 1, then\ p_{cbt}^{rx} \geq \gamma$)

4. Throughput Constraint
   The assigned base station must offer a throughput of at least 7 Mbps, computed via Shannon's formula:

$$Thr_{cbt} = f_b.\log_2 1 + \frac{1 + 10^{(p_{cbt}^{rx} - N)/10}}{1})$$

$$y_{cbt}.Thr_{cbt} \geq y_{cbt}.7 \qquad \forall c, b, t$$

This MILP formulation ensures:

- All TPs are covered
- All throughput constraints are satisfied
- The overall cost is minimized

Results:

1. Deployment Layouts by Scenario

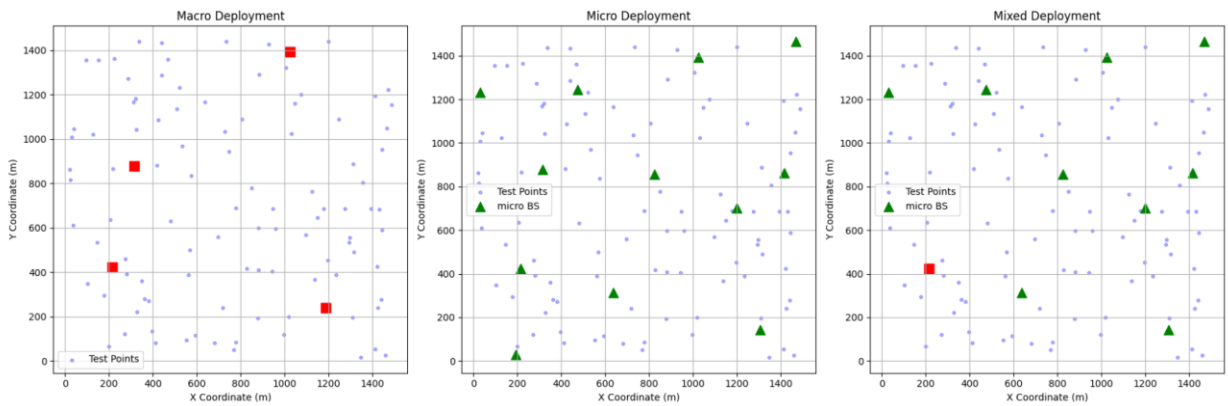

Figure 1

Figure 1 shows the selected base station locations under each scenario:

- Left – Macro Deployment:
  Few macro BSs (red) cover the entire area, with high power but high cost.
- Middle – Micro Deployment:
  More micro BSs (green) are needed due to lower range, but they are cheaper.
- Right – Mixed Deployment:
  A mix of 1 macro and 9 micro BSs achieves optimal cost-performance balance.

2. Cost and BS count comparison

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=== Scenario Comparison ===
Scenario    | Total Cost ($)  | Macro BS | Micro BS | Total BS
-----------------------------------------------------------
macro       | 7.99            | 4        | 0        | 4
micro       | 4.77            | 0        | 12       | 12
mixed       | 4.58            | 1        | 9        | 10

Optimal scenario is 'mixed' with total cost $4.58
Cost savings compared to:
- macro: 42.7% savings
- micro: 3.9% savings
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Figure 2

Despite micro BSs being individually cheaper, combining both types gives the best overall cost-efficiency while using fewer total BSs than micro-only.

3. Sensitivity Analysis

	gamma	thr_min (Mbps)	macro_cost	macro_bs	micro_cost	micro_bs	\
0	-80	5.0	3.48	2	1.90	7	
1	-80	7.0	3.48	2	1.90	7	
2	-80	10.0	3.48	2	1.90	7	
3	-75	5.0	7.99	4	4.77	12	
4	-75	7.0	7.99	4	4.77	12	
5	-75	10.0	7.99	4	4.77	12	
6	-70	5.0	15.66	7	4.70	9	
7	-70	7.0	15.66	7	4.70	9	
8	-70	10.0	15.66	7	4.70	9	

	mixed_cost	mixed_bs
0	1.90	7
1	1.90	7
2	1.90	7
3	4.58	10
4	4.58	10
5	4.58	10
6	10.88	12
7	10.88	12
8	10.88	12

Figure 3

Figure 3 shows how network cost and BS counts change under different conditions:

Parameter	Value Range
Receiver Sensitivity (γ)	from -80 to -70 dBm
Throughput Requirement	from 5 to 10 Mbps

Table 4

Observations:

As γ becomes more strict (from -80 to -70 dBm), the number of required BSs and cost increase across all scenarios.
Higher throughput demands (10 Mbps vs. 5 Mbps) also raise deployment cost.

Mixed deployment consistently offers the best or near-best performance-cost tradeoff.

Answers to the Big Telco Questions:

1. Which is better: macro or micro BSs?
   Micro BSs are significantly more cost-effective while still satisfying all coverage and performance requirements. In this instance, they cost ~80% less than macro BSs.

2. What is the cost if only one type is installed?

   Macro-only deployment: Cost = 7.99

   Micro-only deployment: Cost = 4.77

   Using only micro BSs clearly minimizes deployment cost.

3. Can we save more by combining both types?
   Yes, combining macro and micro BSs reduces the cost compared to macro-only and micro-only deployments. The mixed scenario yields: Cost = 4.58. It is 42.7% cheaper compared to macro-only and 3.9% cheaper compared to micro-only.
   Even though micro-only is quite cheap, mixed deployment slightly outperforms it in both cost and number of total BSs used.

   https://colab.research.google.com/drive/1QJ5EpcB6GICV32s62zWEnNYolrUrYhsT?usp=sharing